

Joseph Wardle

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WORK EXPERIENCE

Pipeline Technical Director - Student Short Films

October 2025 - Present

Brigham Young University Center for Animation

Provo, UT

- Develop and support **Python/Qt** tools and a **USD** pipeline for **50+ artists** on Linux and Windows; troubleshoot DCC tools and publishing failures, and manage shot and asset data.

Sandwich Kwon Do

- Implemented **immutable, versioned USD publishes** and pinned every layer's version at **Tractor** submission, eliminating flicker caused by assets republished mid-render.
- Built a **Maya previs-to-layout workflow** with dry-run validation, **ShotGrid** shot creation and re-cutting, and versioned layout scene saves.
- Built **one-click Maya publishing** through a headless Houdini USD asset builder shared with Substance Painter.
- Built a shared **Maya/Houdini playblast previewer** with FFmpeg burn-ins and **ShotGrid Version uploads**.
- Established the show's **OCIO** color pipeline, baking **OpenDRT and ACES 2.0** to 65³ LUTs for OCIO 2.3 compatibility across Houdini, Maya, Nuke, and Blender.

Honey Business (contributor to an existing pipeline)

- Built whole-collection **VDB-to-mesh foliage proxies**, switched with USD purposes, for **up to 70× faster viewport performance**.

Web Developer

March 2025 - November 2025

Brigham Young University - Computer Science Department

Provo, UT

- Developed and maintained **Django** web services, REST endpoints, and internal admin UIs for non-technical users.
- Built **GitHub Actions** pipelines for testing, container builds, and deployment; used **Docker Compose** for local development and **Jira** for project tracking.

Audio Engineer

September 2023 - November 2025

Brigham Young University

Provo, UT

- Designed sound systems, mixed live audio, and supported streaming for stadium sports, devotionals, and performances.

PROJECTS

Cenote - GPU Path Tracer and Hydra Render Delegate

github.com/joseph-wardle/cenote

- Built a **Rust/Vulkan** GPU path tracer with **Slang** kernels and a **Hydra 2** render delegate for usdview, husk, and Houdini; one unbiased estimator serves interactive previews and final frames.
- Implemented **OpenPBR** shading with baked energy-compensation tables; verified with white furnace tests and comparisons against **pbrt-v4**.

Javelin - Rigid Body Physics Engine (github.com/joseph-wardle/javelin)

2023 - Present

- Built a **C++23** simulator using named modules, fixed 60 Hz physics on a dedicated thread, BVH broad phase, and a warm-started constraint solver; added an **ImGui/OpenGL** viewer and **Tracy** profiling.

Film Grain Synthesis (github.com/joseph-wardle/film_grain)

August 2025

- Reimplemented Newson et al. (2017) in **Rust** with **Rayon** CPU parallelism and **wgpu** compute; multithreaded CPU is a **median 72× faster** than the reference C implementation (~7,200 configurations, ~56,000 renders; median of three runs each).

EDUCATION

Brigham Young University

Provo, UT

Bachelor of Science, Computer Science (Animation & Games Emphasis)

August 2023 - Expected May 2027

- **GPA: 3.95**. Relevant coursework: Data Structures, Multithreading, Linear Algebra, Modeling, Rigging, Shading, FX.

SKILLS

- **Programming:** Python, C++23, Rust, MEL, TypeScript, Bash
- **Pipeline:** USD, ShotGrid, Pixar Tractor, OCIO, RenderMan, MoonRay
- **DCCs & UI:** Maya, Houdini, Nuke, Substance Painter, Blender, Qt/PySide2, ImGui
- **Development:** Git, Perforce, GitHub Actions, Jira, Django, Docker; Linux, Windows